1.2 Analytic Techniques for Finding Limits of Functions

Name _____ Period ____ Date _____

AP Calculus 1.2 (30-minute plan) · Student sheet

I can…

- use the properties of limits and direct substitution.
- find a limit of the form 0/0 by factoring or by using a conjugate.
- find the limit of a difference quotient.

Vocabulary · fill in the missing word

Direct substitution — for a _____ P, the limit of P(x) as $x \to c$ is P(c); the same is true for a rational function if the denominator is not 0 at c

Indeterminate form 0/0 — the result of _____ that gives no information: more work is needed

Conjugate — _____

Part 1 · Notes and examples (fill in each note, then work the example)

Properties of limits: the limit of a sum, difference, product, or constant multiple is the sum, difference, product, or constant multiple of the _____.

Example 1

Find each limit.

a. as $x \to 3$: $2x^2 - x + 4$

b. as $x \to -2$: $(x - 1)(x + 4)$

Rational functions: the limit of R(x) as $x \to c$ is R(c) if the _____ is not 0 at c.

0/0: factor the numerator and the denominator, _____ the common factor, then substitute.

Example 2

Find the limit as $x \to 3$: $\dfrac{x^2 - 9}{x^2 - 3x}$

Radicals: multiply the numerator and the denominator by the _____.

Example 3

Find the limit as $x \to 4$: $\dfrac{\sqrt{x} - 2}{x - 4}$

Part 2 · Partner practice

Partner practice

(AP® Practice) The limit as $x \to 5$ of $\dfrac{x^2 - 25}{x - 5}$ is

(A) 0 (B) 5 (C) 10 (D) nonexistent